

The Market is Huge! Revisiting the Big Market Delusion



ASWATH DAMODARAN

DEC 30, 2019



For the high-profile IPOs that have reached the market in 2019, with apologies to Charles Dickens for stealing and mangling his words, it has been the best and the worst of years. On the one hand, you have seen companies like Uber and Slack, each less than a decade old, trading at market capitalizations in the tens of billions of dollars, while working on unformed business models and reporting losses. On the other, many of these new listings have not only had disappointing openings, but have seen their market prices drop in the months after. In September 2019, we did see an implosion in the value of WeWork, another company that started the listing process with lots of promise and a pricing to match, but melted down from a combination of self-inflicted wounds and public market scrutiny. While these companies were very different in their business models (or lack of them), they shared one thing in common. When asked to justify their high pricing, they all pointed to how big the potential markets for their products/services were, captured in their assessments of market size. Uber estimated its total accessible market (TAM) to be in excess of \$ 6 trillion, Slack's judgment was that it had 5 million plus prospective clients across the world and WeWork's argument was that the commercial real estate market was massive. In short, they were telling big market stories, just as PC makers were in the 1980s, dot com firms in the 1990s and social media companies a decade later. In this post, we will try to concede the allure of big markets, but argue that the allure can lead to delusional pricing. (This post is a not-so short summary of a paper that Brad Cornell and I have written on this topic. You can [find it by clicking here.](#))

The Ingredients

There is nothing more exciting for a nascent business than the perceived presence of a big market for its products and services, and the attraction is easy to understand.

the minds of entrepreneurs in these markets, big markets offer the promise of easily scalable revenues, which if coupled with profitability, can translate into large profits and high valuations. While this expectation is not unreasonable, overconfidence on the part of business founders and their capital providers can lead to unrealistic judgments of future profits, and overly high estimates for what they think their companies are worth, in what I will term the “big market delusion”. That initial overpricing is a common feature of these markets, but results in an inevitable correction that brings the pricing back to earth. In fact, there are three pieces to this puzzle, and it is when they all come together that you see the most egregious manifestations of the delusion.

1. **Big Market:** It is the promise of a big market that starts the process rolling, whether it be eCommerce in the 1990s, online advertising between 2010 and 2018 or artificial intelligence today. In each case, the logic of an impending change was impeccable, but the extrapolation that the change would lead to the creation of huge and profitable markets was made casually. That extrapolation was then used to justify high pricing for every company in the space, with little effort put into separating winners from losers and good from bad business models.
2. **Overconfidence:** Daniel Kahneman, whose pioneering work with Amos Tversky gave rise to behavioral finance as a discipline, described overconfidence as the mother of all behavioral biases, for three reasons. First, it is ubiquitous, since it seems to be present in an overwhelming proportion of human beings. Second, overconfidence gives teeth to, and augments, all other biases, such as anchoring and framing. Finally, there is reason to believe that overconfidence is rooted in evolutionary biology and thus cannot be easily countered. The problem gets worse with big markets, because of a selection bias, since these markets attract entrepreneurs and venture capitalists, who tend to be among the most overconfident amongst us. Big markets attract entrepreneurs, overconfident that their offerings will be winners in these markets, and venture capitalists, overconfident in their capacity to pick the winners.
3. **Pricing Game:** We will not bore you with another extended discourse on the difference between value and price, but suffice to say that young companies tend to be priced, not valued, and often on raw metrics (users, subscribers, revenue).

As a consequence, there is no attempt made to flesh out the "huge market" argument, effectively removing any possibility that entrepreneurs or the venture capitalists funding them will be confronted with the implausibility of their assumptions.

The end result is that young companies in big markets will operate in bubbles of overconfidence, leading them to over estimate their chances of succeeding, the revenues they will generate if they do and how much profit they can generate on revenues:

This does not mean that every company in the big market space will be over priced since a few will succeed and exploit the big market to full effect, but it does mean the companies will be collectively over priced. As is always the case with markets, there will be a time of reckoning, where investors and managers will wake up to the reality that the big market is not big enough to accommodate all their growth dreams and there will be a correction. In the aftermath, there will be finger-wagging and talk of "never again", but the process will be repeated, albeit in a different form, with the next big market.

Case Studies We will not claim originality here, since the big market delusion has always been part of market landscapes, and big markets have always attracted overconfident start ups and investors, creating cycles of bubble and bust. In this section, we will highlight three high profile examples:

1. Internet Retail in 1999 The Big Market: As the internet developed and became accessible to the public in the 1990s, the promise of eCommerce attracted a wave of innovators, from Amazon in online retail in 1994 to Ebay in auctions in 1995, and innovation was aided by the arrival of Netscape Navigator's browser, opening up the internet to retail consumers and PayPal, facilitating online payments. New businesses were started to take advantage of this growing market with the entrepreneurs using the promise of big market potential to raise money from venture capitalists, who attached sky-high prices to these companies. By the end of 1999, not only was venture capital flowing in record amounts to young ventures, but 39% of all venture capital

going into internet companies. The Pricing Delusion: The enthusiasm that entrepreneurs and venture capitalists were bringing to online retail companies sent them into public markets, and as public market interest climbed, many young companies found that they could bypass the traditional venture capital route to success and go directly to public listings. Many of the online retail companies that were listed on public markets in the late 1990s had the characteristics of nascent businesses, with small revenues, unformed business models and large losses, but all of these shortcomings were overwhelmed by the perception of the size of the eCommerce market. In 1999 alone, there were 295 initial public offerings of internet stocks, representing more than 60% of all initial public offerings that year. One measure of the success of these dot.com stocks is that data services created indices to track them. The Bloomberg Internet index was initiated on December 31, 1998, with a hundred young internet companies in it, and it rose 250% in the following year, reaching a peak market capitalization of \$2.9 trillion in early 2000. Because the collective revenue of these companies was a fraction of that value, and most of them were losing money, the only way you could justify these market capitalizations was with a combination of very high anticipated revenue growth accompanied by healthy profit margins in the future, premised on successful entry into a big market. The Correction: The rise of internet stocks was dizzying, in terms of the speed of ascent, but its descent was more precipitous. The date the bubble burst can be debated, but the NASDAQ, dominated in 2000 by young internet companies, peaked on March 10, 2000, and a few months after, the pricing unraveled as shown in the collapse of the Bloomberg Internet Index:

The Bloomberg Internet Index Of the dozens of publicly traded retail companies that existed in March 2000, more than two-thirds failed, as they ran out of cash (and capital access) and their business models imploded. Even those that survived, like Amazon, faced carnage, losing 90% of their value, and flirting with the possibility of shutting down.

2. Online Advertising in 2015 & 2019 The Big Market: The same internet that gave birth to the dot com boom in the nineties also opened the door to digital advertising and while it was slow to find its footing, the arrival of search engines like Yahoo! and Google fueled its growth. The advent of social media altered the game even more

businesses realized that not only were they more likely to reach customers on social media sites, but that social media companies also brought in data about their use that would allow for more focused and effective advertising. The net result of all innovations was that digital advertising grew in the decade from 2005 to 2015, both in absolute numbers and as a percent of total advertising:

As digital advertising grew, firms that sought a piece of this space also entered the market and were generally rewarded with infusions of capital from both private and public market investors. The Pricing Delusion: In a [post in 2015](#), I looked at how the size of the online advertising market skewed the valuations of companies in this market, by looking at publicly traded companies in the space and backing out from market capitalizations what revenues would have to be in 2025, for investors to be even. To do this, I made assumptions about the rest of the variables required to conduct a DCF valuation (the cost of capital, target operating margin and sales to capital ratio) and held them fixed, while I varied the revenue growth rate until I arrived at the current market capitalization. With Facebook in August 2015, for instance, here is what I estimated:

Put simply, for Facebook's market capitalization in 2015 to be justified, its revenues would have to rise to \$129,318 million in 2025, with 93% of those revenues coming from online advertising. Repeating this process for all publicly traded online advertising companies in August 2015:

Imputed Revenues in 2025 in millions of US \$

The total future revenues for all the companies on the list totals \$523 billion. Note that this list is not comprehensive, because it excludes some smaller companies that also generate revenues from online advertising and the not-inconsiderable secondary revenues from online advertising, generated by firms in other businesses (such as Apple). It also does not include the online advertising revenues being impounded in the valuations of private businesses like Snapchat, that were waiting in the wings in 2015. Consequently, we are understating the imputed online advertising revenue that was being priced into the market at that time. In 2014, the total advertising market

globally was about \$545 billion, with \$138 billion from digital (online) advertising. Even with optimistic assumptions about the growth in total advertising and the digital advertising portion of it climbing to 50% of revenues, the total online advertising market in 2025 comes to \$466 billion. The imputed revenues from the publicly traded companies in August 2015 alone exceeds that number, implying that the companies were being overpriced relative to the market (online advertising) from which their revenues were derived.

The Correction? The online ad market has not had a precipitous fall from the heights of 2015, but it has matured. By 2019, not only had investors learned more about the publicly traded companies in the online advertising business, but online advertising has matured. Using the same process that we used in 2015, we imputed revenues for 2029 using data up through November 2019. Those calculations are presented in the table below:

Imputed Revenues in 2029 in millions of US \$

There are signs that the market has moderated since 2015. First, the number of companies shrank, as some were acquired, some failed, and a few consolidated. Second, the market capitalizations had been recalibrated and starting revenues in 2015 are much greater than they were in 2015. As a result, the breakeven revenue in 2029 is \$573 billion, only slightly higher than the imputed revenues from the 2015 calculations despite being four years further into the future. This suggests that the market is starting to take account of the limits imposed by the size of the underlying market. Third, more of the companies on the list have had moments of reckoning with the market, where they have been asked to show pathways to profitability and not just growth numbers. Two examples are Snap and Twitter. For both companies the market capitalizations have languished because of the perception that their pathways to profitability are rocky. In short, if there is a correction occurring in this market, it seems to be happening in slow motion.

3. Cannabis in October 2018

The Big Market: Until recently, cannabis, in any of its forms, was illegal in every country in the United States in most of the world, but that is changing rapidly. By October 2018, smoking marijuana recreationally and medical marijuana were both legal in 10 states, and medical marijuana alone in another 20 states. Outside the United States, much of Europe has always taken a more sanguine view of cannabis, and on October 17, 2018, Canada became the second country (after Uruguay) to legalize the recreational use of the product. In conjunction with this development, new companies were entering the market, hoping to take advantage of what they saw as a “big” market, and excited investors were rewarding them with large market capitalizations. The widespread view as of October 2018 was that the cannabis market would be big, in terms of users and revenues. There were concerns that many recreational cannabis users would continue to use the cheaper, illegal version over the regulated but more expensive one, and that US federal law would be slow to change its view of legality. In spite of these caveats, there remained optimism about growth in this market, with the more conservative forecasters predicting that global revenues from marijuana sales will increase to \$70 billion in 2024, triple the estimated sales in 2018 and the more daring ones predicting close to \$150 billion in sales.

The Pricing Delusion: In October 2018, the cannabis market was young and evolving, with Canadian legalization drawing more firms into the business. While many of the firms were small, with little revenue and big operating losses, and most were privately owned, a few of these companies had public listings, primarily on the Canadian market. The table below lists the top ten cannabis companies as of October 14, 2018, with the market capitalizations of each one, in conjunction with each company’s operating numbers (revenues and operating income/losses, in millions of US \$).

Cannabis Stocks on Oct 14, 2018 (\$ values in millions of US\$)

Note that the most valuable company on the list was Tilray with a market cap of \$13 billion. Tilray had gone public a few months prior, with revenues that barely register (\$28 million) and nearly equal operating losses, but had made the news right after its IPO, with its stock price increasing ten-fold in the following weeks, before subsequently losing almost half of its value in the following weeks. Canopy Growth, the largest and most established company on the list, had the highest revenues at

million. More generally, all of them trade at astronomical multiples of book value, a collective market cap in excess of \$48 billion, more than 20 times collective revenue and 10 times book value. For each company, the high market capitalization relative to any measure of fundamental value was justified using the same rationale, namely the cannabis market was big, allowing for huge potential growth.

The Correction: In the case of the cannabis market, the overreach on the part of both businesses and their investors caught up with them. By October 2019, the assumptions regarding growth and profitability were being universally scaled back, business models were being questioned, and investors were reassessing the pricing of these companies. The best way to see the adjustment is to look at the performance of the major cannabis exchange-traded fund, ETFMG, over the period depicted in the figure below:

Note that within a period of approximately one year, cannabis stocks lost more than 50% of their aggregate value. The damage cut across the board. Tilray and Canopy Growth, the two largest market capitalization companies in the October 2019 snapshot, saw their market capitalizations decline by 80.7% and 38.6% respectively. Given that there was no significant shift in fundamentals, the apparent explanation is that investors came to realize that the “big market” was not going to deliver the previously expected growth rates or the profitability for the expanding group of individual companies.

Common Elements

The three examples that we listed are in very different businesses and have different market settings. That said, there are some common elements that you see in all three and will in any big market setting:

1. **Big Market stories:** In every big market delusion, there is one shared feature. When asked to justify the pricing of a company in the market, especially you see with companies with little to show in terms of fundamentals, entrepreneurs, managers, and investors almost always point to macro potential, i.e., that the retail or medical or advertising or cannabis markets were huge. The interesting aspect is that they rarely express the need to go beyond that justification, by explaining why the specific company they were recommending was positioned to take advantage

that growth. In recent years, the big markets have gone from just words to numbers, as young companies point to big total accessible markets (TAM), seeking higher pricing, often adopting nonsensical notions of what accessible means to get to large numbers.

2. Blindness to competition: When the big market delusion is in force, entrepreneurs, managers and investors generally downplay existing competition, thus failing to factor in the reality that growth will have to be shared with both existing and potential new entrants. With cannabis stocks in late 2018, much of the pricing optimism was driven by the size of the potential market in the United States, assuming legalization, but very few entrepreneurs, managers and investors seemed to consider the likelihood that legalization would attract new players to the market and that illegal sources of supply would maintain their hold on the market.
3. All about growth: When enthusiasm about growth is at its peak, companies focus on growth, often putting business models to the side or even ignoring them completely. That was true in all three of our case studies. With internet stock companies typically based their entire pricing pitch on how quickly they were growing. With social media companies, it took an even rawer form, with growth in users and subscribers being the calling cards for higher pricing. Investors, both private and public, not only went along with the pitch but [often actively encouraged companies to emphasize growth](#) at the expense of profits.
4. Disconnect from fundamentals: If you combine a focus on growth as the basis for pricing with an absence of concern at these companies about business models, you get pricing that is disconnected from the fundamentals. In all three case studies presented in this paper, at the peak of the pricing run up, most of the stocks in each group had negative earnings (making earnings multiples not meaningful), little to show in assets (making book value multiples difficult to work with) and were traded at huge multiples of revenues. Put simply, the pricing was losing its moorings in value, but investors who look at only multiples miss the disconnect.

The one area where the three case studies diverge is in how the pricing delusion corrects itself. For instance, the dot com bubble hit a wall in March 2000 and burst.

few months, as public markets corrected first, followed by private markets, but the question of why it happened at the time that it did remains a mystery. The online advertising run-up has moderated much more gradually over a few years, and if the trend continues, the correction in this market may be smooth enough that investors will not call it a correction. With cannabis stocks, the rise and fall were both precipitous, with the stocks tripling over a few months and losing that rise in the few months.

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Implications

If the big market delusion is a feature of big markets, destined to repeat over time, it behooves us as entrepreneurs, managers, investors and regulators to recognize the reality and modify our behavior.

1. Entrepreneurs and Venture Capitalists

The obvious advice that can be offered to entrepreneurs and venture capitalists, to counter the big market delusion, is to be less over confident, but given that it is not only part of their make up but the driver for exploiting the big market, it will have little effect. Our suggestions are more modest. First, testing out the plausibility of market size assumptions and the viability of the business model you plan to use to exploit the market on people, whose opinion you value but don't operate in your bubble, is a sensible first step. Second, when you get results from your initial business forays that run counter to what you expected to see, don't be quick to rationalize away as aberrations. By keeping the feedback loop open, you may be able to improve your business model and adjust your expectations sooner, to reflect reality. Third, build in safety buffers into your model, allowing you to keep operating even if cash dries up (as it inevitably will when the correction arrives), by accumulating cash and avoiding cost commitments that lock you in, like debt and long term cost contracts. Finally, while you may be intent on delivering the metrics that are priced highly, as users or subscribers, pay attention to building a business model that will work delivering profits, and if forced to pick between the two objectives, pick the latter.

2. Public Market Investors

The big market delusion almost never stays confined to private markets and soon later, the companies in the space list on public markets and are often priced in the markets, at least initially, like they were in private markets. While a risk averse investor may feel it prudent to entirely avoid these stocks, there are opportunities that can be exploited:

- **Momentum investors/traders:** The big market delusion is one explanation for the momentum of young, growth stocks. When fascination with a big market like “transportation” takes hold, it can produce momentum in the prices of innovative companies in that space such as Uber and Lyft, and significant profits along the way. The risk, of course, is that the big market delusion fades and the market corrects as has happened in the case of both Uber and Lyft. As we have emphasized, however, there appears to be no way to time such corrections.
- **Value investors:** The obvious advice is to avoid young, growth stocks whose prices are based on big market stories. But that carries its own risk. In the twelve year stretch beginning in 2007, growth stocks have dramatically outperformed value stocks. As one example, during this period the Russell 1000 growth index outperformed the Russell 1000 value index by an astonishing 4.3% per year. That outperformance was driven in part by stories regarding how technology companies were going to disrupt or invent big markets from housing to entertainment to automobiles. There is a riskier, higher payoff, strategy. Since the big market delusion leads to a collective over pricing, value investors can bet against a basket of stocks (sell short on an ETF like the ETFMG) and hope that the correction occurs soon enough to reap rewards.

In sum, though, young companies make markets interesting and by making them interesting, they increase liquidity and trading.

3. Governments and Market Regulators

In the aftermath of every correction, there are many who look back at the bubble as an example of irrational exuberance. A few have gone further and argued that such episodes are bad for markets, and suggested fixes, some disclosure-related and some putting restrictions on investors and companies. In fact, in the aftermath of every

bursting bubble, you hear talk of how more disclosure and regulations will prevent the next bubble. After three centuries of futility, where the regulations passed in response to one bubble often are at the heart of the next one, you would think that we would learn, but we don't. In fact, over confidence will overwhelm almost every regulatory and disclosure barrier that you can throw up. We also believe that these critics are missing the point. Not only are bubbles part and parcel of markets, they are not necessarily a negative. The dot com bubble changed the way we live, altering not only how we shop but how we travel, plan and communicate with each other. What is more, some of the best performing companies of the last two decades emerged from the debris. Amazon.com, a poster child for dot com excess, survived the collapse and became a company with a trillion-dollar market capitalization. Our policy advice to politicians, regulators and investors then is to stop trying to make bubbles go away. In our view, requiring more disclosure, regulating trading and legislating moderation are never going to stop human beings from overreaching. The enthusiasm for big markets may lead to added price volatility, but it is also a spur for innovation, and the benefits of that innovation, in our view, outweigh the costs of the volatility. We would choose the chaos of bubbles, and the change that they create, over a world run by actuaries where we would still be living in caves, weighing the probabilities of whether firing a good invention or not.

Conclusion

Overconfident in their own abilities, entrepreneurs and venture capitalists are naturally drawn to big markets which offer companies the possibility of huge valuations if they can effectively exploit them. And there are always examples of immense successes, like Amazon, to fuel the fire. This leads to a big market delusion resulting in too many new companies being founded to take advantage of big markets, each company being overpriced by its cluster of founders and venture capitalists. Overconfidence then feeds into public markets, where investors get their cues on valuation and relevant metrics from private market investors, leading to inflated values in public markets. This results in eventual corrections as the evidence accumulates that growth has to be shared and profitability may be difficult to achieve in a competitive environment. This post is a long one, but if you find it interesting, [Brad Cornell](#)

[have a paper expounding a more complete picture here](#). As always, your feedback appreciated!

Paper on the big market delusion

1. [The Big Market Delusion: Valuation and Investment Implications](#)

Previous posts relating to the big market delusion

1. [Big Markets, Over Confidence and the Macro Delusion \(October 2015\)](#)
2. [High and Higher: The Money in Marijuana \(October 2018\)](#)

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